Project Report

Project Title: CAMPUS RECRUITMENT SYSTEM

Introduction

A Campus Recruitment System is a software application that automates the recruitment process between students, companies, and the college administration. It provides a centralized platform where students can register, create profiles, apply for jobs and companies can post vacancies, review applicants and hire suitable candidates. This system reduces paperwork, saves time, improves communication and ensures efficient record management.

The project is developed in C++ using Object-Oriented Programming (OOP) concepts. It includes modules for Admin, Student, Company, Job and Application Management, with features such as registration, login, job posting, job applications, hiring, report generation and permanent data storage using binary files. OOP concepts like inheritance, polymorphism, encapsulation, abstraction, templates, exception handling and file handling are implemented to make the system modular and efficient.

The project successfully manages the complete campus recruitment process, including registrations, job postings, applications, hiring and report generation. Although it improves recruitment efficiency and maintains persistent records, it has limitations such as a console-based interface, lack of password encryption, no support for multiple administrators and the absence of advanced features like interview scheduling, email notifications, resume uploads and online aptitude tests.

Objectives

The main objective of developing the Campus Recruitment System is to automate the campus placement process and reduce manual work involved in recruitment activities. The system provides an easy and efficient platform where students can register, manage their profiles, upload skills and apply for available jobs, while companies can post vacancies and recruit eligible candidates.

The project is developed mainly for educational purposes to demonstrate the practical implementation of Object-Oriented Programming (OOP) concepts using C++. It also aims to improve programming skills in file handling, exception handling, templates, inheritance, polymorphism and data management while creating a real-life software application.

Applications

The Campus Recruitment System can be used in the following areas:

- Colleges and Universities for campus placement management.
- Placement Cells to manage student and company information.
- Students to create profiles and apply for jobs.
- Companies to post job opportunities and recruit candidates.
- Educational Institutions to maintain recruitment records digitally.
- Small organizations requiring a basic recruitment management system.
- Academic projects for learning Object-Oriented Programming concepts.

Methodology

The Campus Recruitment System follows the Object-Oriented Programming (OOP) methodology. Initially, the system requirements were analyzed by identifying three user roles: Administrator, Company, and Student, and defining their respective functionalities. Based on these requirements, the system was designed using separate classes such as User (Abstract Base Class), Student, Company, Admin, Job, Application, Repository (Template Class) and RecruitmentSystem.

OOP Concepts Used:

1. Class and Object
The entire project is built using multiple classes.

2. Encapsulation

Data members are declared as private/protected, while public member functions provide controlled access.

```
class Student : public User
{
    private:
    string dept;
    float cgpa;
    vector<string> skills;
    bool placed;
```

3. Inheritance

Student, Company and Admin inherit from the User class.

```
class Company : public User
{
```

4. Abstraction

The User class is an abstract class because it contains a pure virtual function.

```
virtual void display() const = 0;
virtual void serialize(ofstream &out) const
```

5. Polymorphism

Runtime polymorphism is implemented using virtual functions.

```
c->display();
```

6. Constructor

Parameterized and default constructors initialize objects.

```
Student() : User(), dept(""),
```

7. Destructor

Destructor automatically saves data before program termination.

```
~RecruitmentSystem() {
    saveAll();
}
```

8. Operator Overloading

Insertion, extraction, and comparison operators are overloaded.

```
bool operator==(const Student &o) const {
    return id == o.id; }
```

9. Friend Function

A friend function generates the complete system report.

```
    friend void generateReport(RecruitmentSystem &sys);
};
```

10. Templates

Generic Repository class is implemented using templates.

```
template <class T>
class Repository
{
```

11. Exception Handling

Custom and standard exceptions improve program reliability.

```
    throw FileError();
```

12. File Handling

Binary files permanently store project data.

```
    item.deserialize(in);
```

13. STL (Standard Template Library)

The project uses: Vector, string, algorithm

Functions:

```
    data.push_back(item);
```

14. Static Members

Static variables automatically generate unique IDs.

```
public:
    static int nextId;
```

Result

The Campus Recruitment System was successfully developed using C++. The system enables students to register, manage profiles, upload skills, apply for jobs and track applications. Companies can post jobs, review applicants and hire eligible candidates, while the administrator manages records and generates recruitment reports.

The project successfully demonstrates the implementation of Object-Oriented Programming concepts, including inheritance, polymorphism, encapsulation, templates, exception handling, operator overloading, STL and binary file handling. The system performs efficiently and ensures permanent data storage with reliable record management.

System Output:

```
==========================================
   CAMPUS RECRUITMENT SYSTEM
==========================================
 1. Admin Login
 2. Company Register
 3. Company Login
 4. Student Register
 5. Student Login
 6. Exit
==========================================
Enter choice:
```

```
Γëà Welcome Tasin!

========== STUDENT DASHBOARD ==========
 1. View Profile
 2. Upload Skills
 3. Edit Details
 4. View Jobs
 5. Apply for Job
 6. View My Applications
 7. Logout
=======================================
Enter choice:
```

```
Enter choice: 3
JobID:3002 | cse | Closed
JobID:3004 | cse | Closed
JobID:3006 | Software | Open
JobID:3008 | Software | Open
JobID:3010 | Software_Engineer | Open

=========== ADMIN PANEL ============
 1. View All Students
 2. View All Companies
 3. View All Jobs
 4. Delete Student
 5. Delete Company
 6. Delete Job
 7. Generate Report
 8. Logout
====================================
Enter choice:
```

Conclusion

In this project, a complete Campus Recruitment System was designed and implemented using C++. The system successfully automates the campus placement process by connecting students, companies and the administrator on a single platform. The project achieved its objectives by reducing manual work, organizing recruitment data efficiently, providing secure login systems, preventing duplicate job applications and maintaining permanent records through binary file handling. It also successfully demonstrates the practical application of various Object-Oriented Programming concepts.

Although the project performs its intended functions successfully, it has some limitations. It is a console-based application without a graphical user interface. Passwords are stored without encryption, interview scheduling and online tests are not supported, resume upload functionality is unavailable and only a single administrator account is implemented. These limitations reduce the system's scalability for real-world deployment.

In the future, this project can be enhanced by developing a graphical user interface (GUI) or web-based version. Additional features such as resume uploading, interview scheduling, email and SMS notifications, online aptitude tests, company verification, secure password encryption, database integration (MySQL), advanced search and filtering, and detailed placement analytics can be incorporated. These improvements will make the system more user-friendly, secure and suitable for real-world campus recruitment management.